underperform by 1.3% after fees. Controlling for more sophisticated risk factors, they find that the returns of mutual funds in excess of their risk benchmarks is only 0.1% before fees, and the corresponding number after fees is –1.0%.

French (2008) goes further in his presidential address to the American Finance Association. He computes the cost of active management to all investors—including individual investors, mutual funds, pension funds and other institutional investors, and hedge funds in the U.S. equity market. He estimates that the average investor would be better off by 0.7% per year switching from active management to index funds.

Even though active managers as a group underperform, do they underperform less (or perhaps slightly add value) during bear markets or recessions? The evidence is mixed. On the one hand, Moskowitz (2000) and Kosowski (2012) find that active mutual funds significantly out-perform during recessions (as defined by the National Bureau of Economic Research). But De Souza and Lynch (2012) find the opposite.

The bottom line is that, with or without adjusting for risk, the typical active mutual fund delivers average underperformance.

Larger Mutual Funds Do Worse

Larger mutual funds do worse than smaller ones, so there are *diseconomies of scale*.[29] As a fund gets larger, it becomes harder for it to take on large positions in stocks. Many alpha opportunities are concentrated in more illiquid segments of the market. Good investment ideas thus are hard to scale. Indeed, large-fund performance is most hampered when funds hold more illiquid stocks. Larger funds also have to employ more analysts, and this makes coordination more difficult. With more employees, it is harder to pass information up the organization and to act on it. (In economics these costs are called *hierarchy costs*.[30])

More Expensive Funds Do Worse

In most industries you get what you pay for. When you shop at Saks Fifth Avenue, you pay more than at the dollar store, but you get better quality. When you eat at a three-star Michelin restaurant, you pay more than at McDonalds, but you get a more delicious meal in elegant surroundings. In so many arenas, the more you pay, the more you get. But in mutual fund land, it's the exact opposite: when you pay more, you get less.[31] Carhart (1997) finds that the more fees you pay to a mutual fund, the *lower* the return. Investors are best served by investing in funds with low expenses. Gruber (1996) finds that top performing funds charge *less* than the

[29] See Chen et al. (2004), Pollet and Wilson (2008), and Yan (2008).

[30] See Stein (2002).

[31] For some theoretical explanations see Christoffersen and Musto (2002) and Gil-Bazo and Ruiz-Verdu (2008). In addition, see the Berk and Green (2004) model, which is discussed in section 3.5.